Report

Of

AI-ML Virtual Internship: Explainable Plant Disease Classification Using Grad-CAM and LIME with ResNet18

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number: CS-2341030

Name of Student: Kiran Shetty

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Kiran Shetty, do hereby declare that the internship report titled "AI-ML Virtual Internship" has been completed by me to fulfil the requirement for the award of the Bachelor of Technology in Computer Science and Engineering. All the references have been quoted, and I have not taken any material from any other source as well. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: Kiran Shetty

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AI-ML Virtual Internship Program. I am thankful to EduSkills Foundation, AICTE, the National Internship Portal, and Google for Developers for providing this opportunity and the resources necessary to complete this internship and project successfully.

I am sincerely grateful for the guidance, constructive feedback, and encouragement received during this internship, which helped me understand and apply key concepts of Artificial Intelligence, Machine Learning, and Explainable AI to a real project.

I also express my gratitude to my parents and friends for their constant support and blessings.

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3. INTERNSHIP COMPLETION CERTIFICATE

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Certificate of Virtual Internship

This is to certify that

Kiran Shetty

IILM University, Greater Noida

has successfully completed the 8-weeks

AI-ML Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : **India Edu Program**
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Student ID: STU67e580e94f2e31743093993

INTERNSHIP GRADE O

GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are among the fastest-growing areas of technology, with applications in healthcare, agriculture, finance, automation and computer vision. Deep learning, a major branch of AI, has made it possible to analyse large amounts of image data and automatically identify patterns that may be difficult for humans to detect.

During my AI-ML Virtual Internship, I gained practical exposure to various concepts related to Artificial Intelligence, Machine Learning, neural networks, and computer vision. The internship provided an opportunity to understand how machine learning models are developed, trained, evaluated, and applied to real-world problems. Along with the learning modules and practical exercises completed during the internship, I developed a project titled "Explainable Plant Disease Classification Using Grad-CAM and LIME with ResNet18."

The main purpose of the project was to classify plant diseases from leaf images using a deep learning model and to make the predictions understandable using Explainable Artificial Intelligence (XAI) techniques. Although deep learning models can achieve high classification accuracy, they are often considered "black box" systems because they do not clearly explain why a particular prediction has been made. Therefore, this project used Grad-CAM and LIME to generate visual explanations for the predictions of the trained model.

A pre-trained ResNet18 model was used with transfer learning to classify plant leaf images into different disease categories. The PlantVillage dataset was used for training and testing the model. The project further compared Grad-CAM and LIME based on explanation quality, robustness, and computational efficiency.

This internship and project helped me gain practical knowledge of deep learning, transfer learning, image classification, model evaluation, Explainable AI, and web-based deployment of machine learning models.

4.2 Organization Profile

The AI-ML Virtual Internship was organized through the support of EduSkills Foundation, AICTE, the National Internship Portal, and Google Developers.

EduSkills Foundation focuses on providing skill-based and industry-oriented learning opportunities to students. It works towards improving practical knowledge and employability by offering training programs and virtual internships in emerging technology domains.

The AI-ML Virtual Internship provided students with an opportunity to gain exposure to Artificial Intelligence and Machine Learning through structured learning modules and practical activities. The program covered important concepts related to neural networks, deep learning, and computer vision.

Google for Developers supported the learning initiative by providing access to relevant technical learning resources. AICTE and the National Internship Portal supported the internship program by connecting students with industry-oriented learning opportunities.

During the internship, I gained practical knowledge of AI and ML concepts and applied these concepts to the development of an explainable plant disease classification system.

4.3 Problem Statement

Plant diseases can significantly affect crop quality and agricultural productivity. Early and accurate identification of plant diseases is important for taking appropriate preventive measures. Traditionally, disease identification depends heavily on manual observation by farmers or agricultural experts, which may be time-consuming and difficult when expert assistance is not easily available.

Deep learning models can classify plant diseases from leaf images with high accuracy. However, one major limitation is that these models often behave like black-box systems. A model may predict a particular disease with high confidence, but the user may not know which part of the leaf image influenced that prediction.

This lack of transparency can reduce trust in AI-based agricultural systems. Therefore, there is a need for a system that not only predicts the disease correctly but also provides a visual explanation for the prediction.

The problem addressed in this project was to develop a plant disease classification system using deep learning and to compare two Explainable AI techniques, namely

Grad-CAM and LIME, to determine which method provides more useful, faithful, robust and efficient explanations.

4.4 Project Objectives

The main objectives of the project were:

- To develop a deep learning model for the classification of plant diseases from leaf images.
- To use transfer learning with the ResNet18 model for improving classification performance.
- To classify images into multiple plant disease categories.
- To apply Grad-CAM for visualising the important regions responsible for model predictions.
- To apply LIME for generating local explanations using image superpixels.
- To compare Grad-CAM and LIME based on their explanation of quality.
- To evaluate the faithfulness of the explanations using a pixel-deletion test.
- To analyse the robustness of explanations under Gaussian noise.
- To compare the computational efficiency of Grad-CAM and LIME.
- To develop a user-friendly application for image-based disease prediction and explanation.

4.5 Scope of the Project

The scope of this project is focused on the application of deep learning and Explainable Artificial Intelligence for plant disease classification.

The project uses plant leaf images as input and predicts the disease category using a trained ResNet18-based deep learning model. The system also generates visual explanations to show the regions of the image that influenced the model's prediction. The project includes the following areas:

- Image-based plant disease classification.
- Transfer learning using ResNet18.
- Classification of 16 plant disease categories.
- Use the PlantVillage dataset.

- Application of Grad-CAM and LIME for explainability.
- Comparison of XAI methods using quantitative evaluation.
- Robustness testing using Gaussian noise.
- Comparison of computational efficiency.
- Development of a FastAPI backend and a React-based frontend for deployment.

The project is currently limited to the selected dataset and supported disease classes. Real-world deployment may require additional training using field images captured under different lighting conditions, backgrounds, and image quality levels.

4.6 Technologies and Tools Used

The following technologies and tools were used during the development of the project:

Programming Language: Python, used for data processing, model development, training, and explainability experiments.

Deep Learning Framework: PyTorch, used to implement and fine-tune the ResNet18 deep learning model.

Pre-trained Model: ResNet18. A pre-trained ResNet18 model was used with transfer learning, and the final classification layer was modified for the required plant disease categories.

Dataset: PlantVillage Dataset, containing 20,632 images representing 16 disease classes across three crop species.

Explainable AI Techniques: Grad-CAM and LIME.

Backend: FastAPI, used to create REST APIs for prediction and explanation generation.

Frontend: React, used to build the user interface for uploading plant images and viewing predictions and explanations.

Supporting Tools and Libraries: NumPy, OpenCV, Matplotlib, Scikit-learn, Captum, and Jupyter Notebook / Google Colab / VS Code.

4.7 System Architecture

The system architecture of the project consists of the following major stages:

Input Plant Leaf Image

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Image pre-processing

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ResNet18 Deep Learning Model

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Plant Disease Prediction

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Explainable AI Module (Grad-CAM / LIME)

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Results Displayed to the User

Detailed Working

First, the user uploads a plant leaf image to the system. The image is resized to the required input size and normalised before being passed to the trained ResNet18 model.

The model extracts important visual features from the leaf image and predicts the corresponding disease class. After the prediction is generated, the Explainable AI module produces visual explanations. Grad-CAM generates a heatmap showing the regions of the leaf that were most important for prediction, while LIME divides the image into superpixels and identifies the regions that contribute most to the predicted result.

The final output includes:

- Predicted plant disease.
- Prediction confidence.
- Grad-CAM explanation.
- LIME explanation.
- Comparison of the explainability methods.

4.8 Methodology

Step 1: Dataset Collection

The PlantVillage dataset was used for the project. The dataset contained plant leaf images belonging to different crop species and disease categories.

Step 2: Data pre-processing

The images were resized to 224 × 224 pixels to match the input requirements of ResNet18 and were normalised before training. The dataset was divided into 70% training data, 15% validation data, and 15% testing data. Data augmentation techniques such as image flipping, rotation, brightness adjustment, and scaling were applied to improve the generalisation capability of the model.

Step 3: Model Development

A pre-trained ResNet18 model was used through transfer learning, and the existing classification layer was modified to classify the required plant disease classes. Initially, selected layers were frozen, and later the complete model was fine-tuned to improve performance.

Step 4: Model Training

The model was trained using the training dataset and evaluated using validation data. Cross-entropy loss and the Adam optimizer were used during training. Early stopping and learning-rate scheduling were used to improve the training process and prevent unnecessary training.

Step 5: Model Evaluation

The trained model was evaluated using the test dataset. The evaluation metrics included Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix.

Step 6: Explainability Using Grad-CAM

Grad-CAM was applied to generate heatmaps showing the important regions of the leaf image that contributed to the model's prediction.

Step 7: Explainability Using LIME

LIME was used to generate local explanations. The image was divided into superpixels, and the important regions contributing to the prediction were identified.

Step 8: Comparison of Explainability Methods

The two methods were compared using faithfulness, robustness, and computation time. A pixel-deletion test was used to determine whether removing important pixels caused the model's confidence to decrease, and Gaussian noise was added to test the stability of the explanations.

Step 9: Deployment

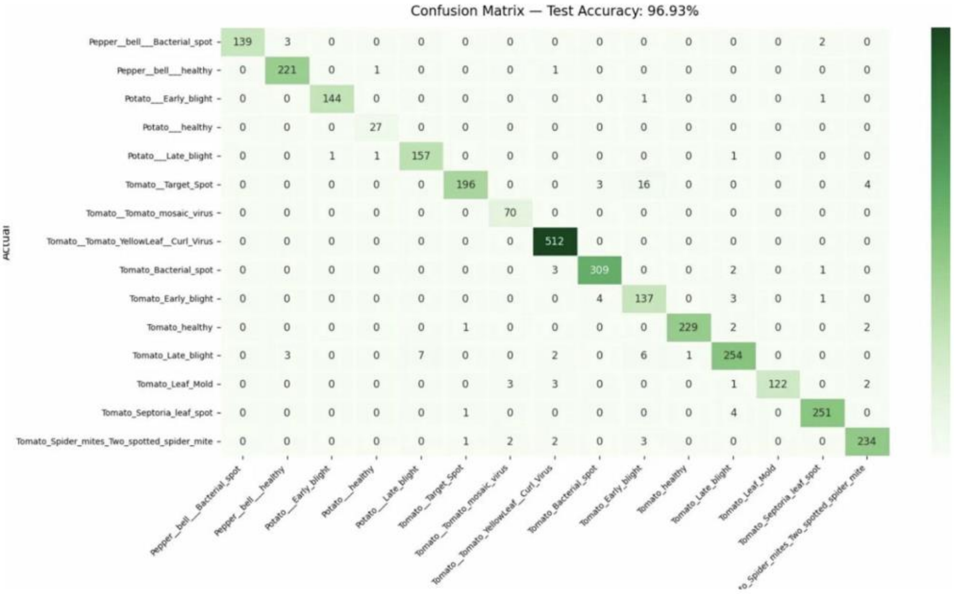
The complete model was integrated into a web application using FastAPI for the backend and React for the frontend. Users could upload a leaf image and view the predicted disease along with the Grad-CAM and LIME explanations.

4.9 Expected Outcomes

The major outcomes achieved through this project were:

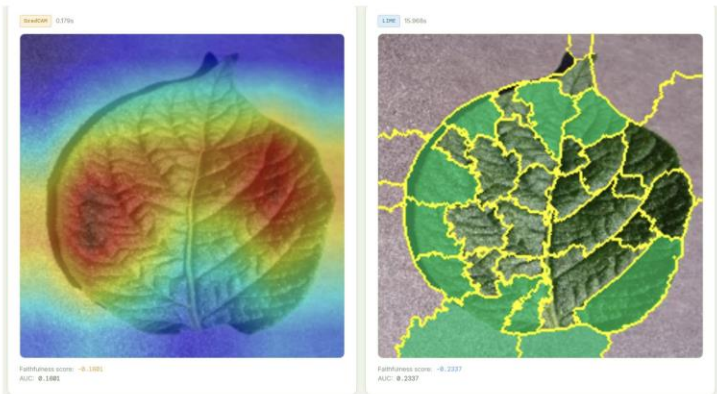
- A deep learning model was successfully developed for plant disease classification.
- The ResNet18 model achieved 96.93% test accuracy, with a macro F1-score of approximately 0.97.
- The model successfully classified plant leaf images into 16 disease categories.
- Grad-CAM successfully generated heatmaps showing important disease-related regions.
- LIME generated local explanations using image superpixels.
- Grad-CAM demonstrated better faithfulness than LIME in the pixel-deletion test, achieving a deletion AUC of 0.31 compared with 0.42 for LIME.
- Grad-CAM was more robust under Gaussian noise, with a mean SSIM of 0.88 compared with 0.64 for LIME.
- Grad-CAM was significantly faster, requiring approximately 0.05 seconds per image, while LIME required approximately 8.2 seconds per image.
- The complete system was successfully deployed using FastAPI and React.

Confusion Matrix



VISUAL COMPARISON

Grad-CAM vs. LIME — Same Image, Same Model



4.10 Certificate of Completion and Supporting Documents

The certificate of successful completion of the AI-ML Virtual Internship has been attached to this report as supporting evidence. The internship was completed through EduSkills Foundation with the support of AICTE, the National Internship Portal, and Google for Developers. The certificate confirms the successful completion of the internship program.

The following documents have been attached along with this report:

- Internship Completion Certificate.

CHAPTER 5

REFERENCES

- Hughes, D. P., & Salathé, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. PlantVillage Dataset.
- Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems (NeurIPS).
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition (ResNet). IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. IEEE International Conference on Computer Vision (ICCV).
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier (LIME). ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
- FastAPI Documentation. Available at: https://fastapi.tiangolo.com/
- React Documentation. Available at: https://react.dev/
- EduSkills Foundation. Available at: https://eduskillsfoundation.org/
- Google for Developers. Available at: https://developers.google.com/
- All India Council for Technical Education (AICTE) – National Internship Portal. Available at: https://internship.aicte-india.org/